

Urban Water Journal

ISSN: 1573-062X (Print) 1744-9006 (Online) Journal homepage: www.tandfonline.com/journals/nurw20

Assessment of the impact of the rise in Lake Victoria water levels on urban flooding using a GIS-based spatial flood modelling approach

Seith N. Mugume, Greson Abasabyoona, Jorem Engwau, Jotham Sempewo, Bas Van de Sande & David Butler

To cite this article: Seith N. Mugume, Greson Abasabyoona, Jorem Engwau, Jotham Sempewo, Bas Van de Sande & David Butler (2024) Assessment of the impact of the rise in Lake Victoria water levels on urban flooding using a GIS-based spatial flood modelling approach, Urban Water Journal, 21:2, 219-233, DOI: 10.1080/1573062X.2023.2284960

To link to this article: https://doi.org/10.1080/1573062X.2023.2284960

View supplementary material

Published online: 12 Dec 2023.

Submit your article to this journal

Article views: 989

View related articles

View Crossmark data

Citing articles: 16 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=nurw20

URBAN WATER JOURNAL
2024, VOL. 21, NO. 2, 219–233
https://doi.org/10.1080/1573062X.2023.2284960

Taylor & Francis
Taylor & Francis Group

RESEARCH ARTICLE

Check for updates

Assessment of the impact of the rise in Lake Victoria water levels on urban flooding using a GIS-based spatial flood modelling approach

Seith N. Mugume[a], Greson Abasabyoona[a], Jorem Engwau[a], Jotham Sempewo[a], Bas Van de Sande[b] and David Butler[c]

[a]Department of Civil and Environmental Engineering Makerere University, Kampala, Uganda; [b]Department of Flood Risk and Urban Water Management, CDR International, Amersfoort, The Netherlands; [c]Centre for Water Systems, Department of Engineering, University of Exeter, Exeter, UK

ABSTRACT

During the period 2018–2021, Lake Victoria experienced a rapid rise in water levels that caused significant urban flooding consequences. However, there are no studies that have extensively investigated the effect of the rising lake water levels on urban flooding. This research responds to this by developing and applying a GIS-based spatial flood modelling approach to simulate the resulting flooding impacts in Entebbe City, Uganda, corresponding to the May 2021 observed maximum lake water level of 13.49 m. Using the validated GIS-based flood model, the simulated flooded area and average flood depth were determined as 9.1 km^2 and 1.52 m, respectively. Furthermore, through application of GIS overlay analysis, the total number of affected buildings and length of urban roads were quantified. The proposed methodology provides a robust and computationally efficient approach for identification of at-risk buildings along the lake shoreline and for testing the effectiveness of potential adaptation strategies.

ARTICLE HISTORY
Received 9 January 2023
Accepted 8 November 2023

KEYWORDS
Flood modelling; flood resilience; Lake Victoria; water level rise

1. Introduction

Globally, the risk of urban flooding has been exacerbated by climate change, rapid urbanisation and changing demographic patterns in recent years (Hirabayashi et al. 2021; Huang et al. 2020; Tellman et al. 2021; UNDRR 2020). Between 2019 and 2021, Lake Victoria in east Africa experienced a rapid rise in water levels that caused significant urban flooding impacts and consequences that included damage to properties, roads and jetties, hydropower plants and displacement of an estimated 200,000 people (Khaki and Awange 2021; Mafaranga 2020; MWE 2020; NASA 2020; NBI 2020). The observed May 2021 maximum lake level of 13.49 m, which exceeded the 1964 historical maximum level of 13.41 m, represented a rapid increase of 15.4% (1.8 m) within a relatively short period of 3.2 years (MWE 2020).

It should be noted that Lake Victoria's water levels are largely influenced by rainfall, with 80% of its refill coming from direct rainfall on the lake surface, and only 20% from basin recharge (Khaki and Awange 2021). The previous maximum lake level rises which occurred in 1878, 1964 and 1998 were largely attributed to rainfall variability, increased tributary inflows and sub-optimal regulation of lake outflows at Jinja (Funk et al. 2015; Inne, Van Lipzig, and Thiery 2018b; MWE 2020; Tate et al. 2004; Wardlaw, Sharif, and Kimaite 2005; Wilberforce et al. 2022).

However, recent studies have attributed the rising lake water levels to climate change, rapid population growth and urbanisation. Mary et al. (2014) and Olaka et al., (2019) applied statistical downscaling of ensembles of global climate model (GCM) data and projected an increase in annual rainfall and extreme daily rainfall intensities over the lake basin of between 5–25% and up to 40% by 2100. Furthermore, the population within the Lake Victoria basin increased by 28.6% from 35 million in 2006 to 45 million 2017 (Güneralp et al. 2018). On the other hand, a drastic increase in built-up areas of 800% occurred between 1985 and 2014 (Mugo et al. 2020), leading to intensive property developments in low lying flood prone areas, increased water pollution and flooding exposure, particularly in cities and urban areas located on the shoreline (NASA 2020, 2021; World Bank 2018). This therefore underscores the need for research effort focused on developing new approaches for quantitative assessment of lake water level induced flooding impacts and consequences in cities.

In recent work, a combination of GIS-based flood modelling and Multiple Criteria Decision Analysis (MCDA) techniques were applied to assess pluvial flood risk in the Attica Region, Greece (Feloni, Mousadis, and Baltas 2020), fluvial flood risk in the Bowen River Catchment in central Queensland, Australia (Chen et al. 2021) and coastal flood risk in south eastern China (Tang et al. 2021). Furthermore, Ntajal et al. (2017) applied a GIS-based weighted overlay analysis approach to develop flood hazard maps for the Lower Mono River Basin in west Africa. This study did not use flood modelling, but utilised expert judgement to attribute flood risk to various factors such as soil type, land use, topography and rainfall.

However, to the authors' knowledge, there are no studies that have extensively investigated the effect of rising lake water levels on urban flooding in the Lake Victoria basin. This study

CONTACT Seith N. Mugume smugume@gmail.com
Supplemental data for this article can be accessed https://doi.org/10.1080/1573062X.2023.2284960
© 2023 Informa UK Limited, trading as Taylor & Francis Group

therefore was focused on development and application of a methodology that combines statistical modelling of the average recurrence interval of the observed May 2021 maximum lake water level rise and GIS-based spatial flood modelling approach for quantitative assessment of the lake water level rise induced flooding in Entebbe City. The resulting flooding impacts and consequences were quantified using the flooded area and flood depths as key flooding indicators. This approach assumes that by modelling the impacts of the maximum lake water level that was recorded in May 2021 on the lake shoreline flooding extent and depths, the emerging impacts of climate change and urbanisation on lake shoreline flooding can be quantified, without attributing the cause of the flooding to a specific threat (e.g. Dawson et al. 2008).

The outputs of this study provide a more reliable approach that can assist decision makers in regional authorities such as the Nile Basin Initiative, the Lake Victoria Basin Commission and national level authorities such water ministries and environmental protection authorities in delineation of flood buffer zones, formulation of lakefront building control policies and testing the effectiveness of potential adaptation strategies for enhancing flood resilience in lake shoreline urban areas.

2. Methods

2.1. Study area and data sets

2.1.1. Lake Victoria catchment, recorded water levels and outflows

Lake Victoria, the second largest freshwater lake in the world, with a surface area of 68,800 km^2, is one of the major sources of the Nile River. The Lake is shared by Uganda (43%), Kenya (6%) and Tanzania (51%) and is fed by 23 major rivers, with Kagera river being the largest. The Lake Victoria basin has a catchment area of 183,000 km^2 which extends to Rwanda, Burundi, Uganda, Kenya and Tanzania (Inne, Van Lipzig, and Thiery 2018b; NASA 2020; NBI 2020; Wardlaw, Sharif, and Kimaite 2005). The hydrology of Lake Victoria is complex and has been the subject of various studies focused on water balance modelling considering current and potential future climate change impacts and outflow regulation (Inne, Van Lipzig, and Thiery 2018a, 2018b; Kizza 2012; Mary, Willems, and Niwagaba 2014; Phoon, Shamseldin, and Vairavamoorthy 2004; Sutcliffe and Parks 1999; Tate et al. 2004; Wardlaw, Sharif, and Kimaite 2005).

Specifically, the water level variations are influenced by a multiplicity of factors that are determined by the lake water balance described in Equation 1 (Herrnegger et al. 2021; Inne, Van Lipzig, and Thiery 2018b; Tate et al. 2004; Yin and Nicholson 1998). These include precipitation on the lake, *P*; inflow from the lake basin tributary rivers, (Q_{in}); evaporation *(E)* and lake outflow (Q_{out}).

$$\frac{dL}{dT} = P - E + \frac{Q_{in-Q_{out}}}{A} \quad (1)$$

Where *dL/dT* is the difference in lake level *dL* (m) per day; *P* is the precipitation over the lake (m/day); E the evaporation of the lake (m/day); Q_{in} the inflow into the lake (m/day) that is contributed by tributary rivers (m^3/d); Q_{out} the lake outflow (m^3/d) and *A* is the surface area of Lake Victoria (6.83×10^{10} m^2). This difference in lake level is added to the lake level of the previous day.

The outflow from the lake, Q_{out} which was originally controlled by the Rippon falls before 1954, is now partially controlled by water released through Owen Falls Dam Complex in Jinja based on the 'agreed curve' that relates the lake water level to the outflow, in such a way that the pre-1954 original lake level/discharge relationship is maintained (Wardlaw, Sharif, and Kimaite 2005; Yin and Nicholson 1998). The rating curve is presented in Equation 2 (Sene 2000).

$$Q_{out} = 66.3(L - 7.96)^{2.01} \quad (2)$$

Where Q_{out} is the outflow in m^3/d from the lake and *L* the Lake Victoria level in m above datum (the mean sea level at Mombasa of 1121 mASL).

2.1.2. Study area

Entebbe City, located in Wakiso District on a peninsula projecting into Lake Victoria, approximately 40 km southwest of Kampala, has traditionally been a centre of commerce and trade, tourism and aviation (Figure 1). It comprises of urban, peri-urban, rural, and fishing communities with a population of approximately 70,000 people, an area of 35.58 sq. km of land and approximately 27 km of coastline (Hoffman 2021; UBOS 2016). The recent rise in Lake Victoria water levels led to the flooding of many residential buildings and infrastructure in Entebbe City. Figure 2 presents a picture of flooded residential houses located near the north western shoreline, while Figure 3 presents a picture of a perimeter wall located on the south western shoreline of Entebbe whose foundation was eroded by wave action resulting from the increased lake water levels.

2.1.3. Data sets used in the study

Table 1 presents the various types of data sets that were utilised in the study.

2.2. Statistical modelling approach for prediction of extreme lake water levels

2.2.1. Historical water level data analysis

The observed lake water level data recorded at Jinja Pier for a period of 74 years (1948–2021) was obtained from the Directorate of Water Resources Management, Ministry of Water and Environment, Kampala. Additional lake water level data for the period 1900 to 1947 was digitized from previous studies to generate a water level data set of approximately 121 years (Kite 1982; Tate et al. 2004; Yin and Nicholson 1998). The annual maximum series (AMS) method (Cunnane 1973; Madsen, Rasmussen, and Rosbjerg 1997) was applied to derive the annual maximum water levels (AMWLs) for each recorded year to form a data set for statistical modelling.

2.2.2. Statistical modelling methods

Three statistical frequency analysis models, the Generalized Extreme Value (GEV), Gumbel and Log Pearson Type III extreme value distributions, were applied in the study (Lanxin et al.

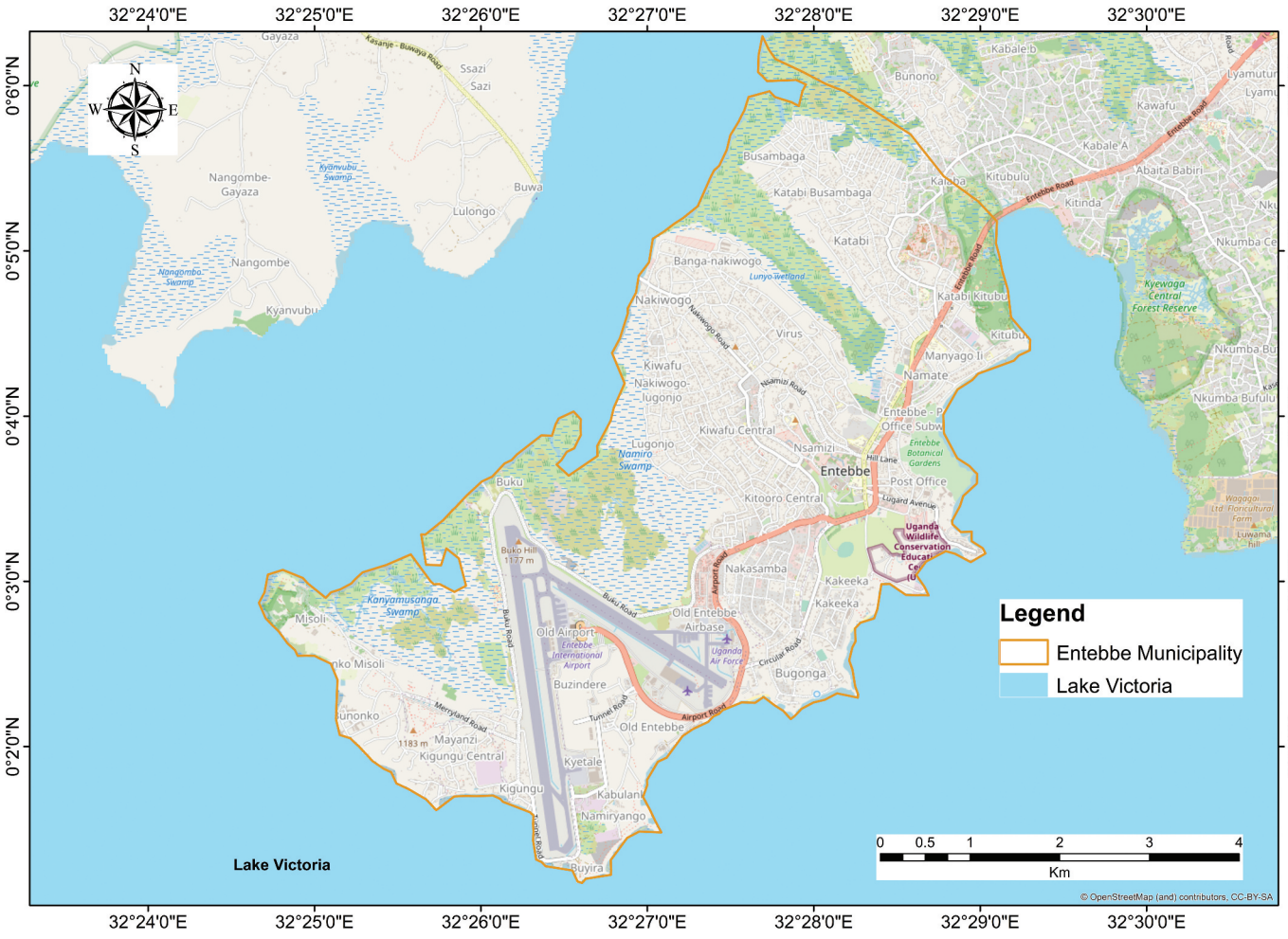


Figure 1. Location of Entebbe City in Uganda.

2020). Data fitting was undertaken using these distributions to predict AMWLs for different return periods, T ($T = 5$, 10, 25, 50, 75, 100 and 200 yrs). The probability distribution functions of the extreme value distributions are described in detail in Supplementary Information Sections 1.1, 1.2 and 1.3.

2.2.3. Comparison between the observed and predicted and AMWLs

The return period, T for the observed data was computed using Equation 3;

$$T_m = (N + 1). \tag{3}$$

Where T_m is the return period corresponding to the mth annual maximum water level x_m, m is the ranking of the water level data from high to low, N is the length of the dataset.

After estimating the parameters for the Gumbel and GEV distributions by the maximum likelihood estimation (MLM) method, the AMWLs corresponding to each return period, T were computed. The procedure for computation of AMWLs is presented in the Supplementary Information Section 1.2.

2.2.4. Assessment of goodness of fit of the statistical models

The main motivation for carrying out statistical modelling with more than one frequency distribution is to select a distribution that provides the best fit of the observed data (Sudong and Huang 2011). The goodness of fit, which provides an estimate of prediction accuracy of a given statistical model was assessed using two parameters i.e. Nash-Sutcliffe Efficiency Index (NSE) and the root mean square error (RMSE) (Gupta and Kling 2011; Seenath, Wilson, and Miller 2016). The NSE values range from negative infinity to 1. NSE values between 0 and 1 are generally acceptable, with 1 as the best value (Gupta and Kling 2011; Hwang, Heon Ham, and Hoon Kim 2012).

$$\mathrm{NSE} = 1 - \frac{\sum_{t=1}^{t_y} \left(x_o^t - x_m^t\right)^2}{\sum_{t=1}^{t_y} \left(x_o^t - \bar{x}_o\right)^2} \tag{4}$$

Where $\bar{x}_o$ is the mean of the observed water levels; x_m the modelled water levels; and x_o^t, the observed water level at time t.

On the other hand, for the RMSE parameter, the lower the value of the RMSE, the higher the accuracy of the prediction (Gupta and Kling 2011; Seenath, Wilson, and Miller 2016).

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{t=1}^{N} \left(x_s^t - x_o^t\right)^2}{N}} \tag{5}$$

Where x_o is the observed water level and x_s the modelled water level and N is the number of observations.



Figure 2. Flooded residential houses located near the north western shoreline.



Figure 3. A section of the perimeter wall on the south eastern shoreline of the lake that failed due to erosion and wave action caused by the rising lake water levels.

2.3. GIS-based spatial flood modelling approach

2.3.1. Flood risk assessment approach

In this study, flood risk was assessed as a function of probability of occurrence of a given flood event and the consequences of occurrence of the flooding event (Gayen, Vallejo Villalta, and Mafizul Haque 2022; Sande Van de, Lansen, and Hoyng 2012). Flood risk, F_r (Equation 5) is therefore a function of the hazard (i.e. frequency of occurrence of flooding of a given magnitude) and the vulnerability (susceptibility) of the receptor exposed to the hazard (Chen et al. 2021).

$$Flood\ risk, F_r = P_{f,T} \times F_c \quad (6)$$

Where F_r is the flood risk, P_f the probability of flooding of a given return period, T and Fc the resulting consequences of flooding.

The probability of flooding was quantified through statistical analysis of observed water level time series, which form the basis for simulation of the flooding impact indicators, that is flooded area (extent) and flood depths. The resulting flooding extent and depths caused by a specific Lake water level rise are influenced by the topography (elevation and slope), soil type, vegetation cover, existing storm water drainage system, proximity of buildings to the lake's shoreline, presence of wetlands which act as flood buffers, presence (or absence) of flood defence walls among others (Chen et al. 2021; Tang et al. 2021). On the contrary, the consequences of flooding are dependent on the population density, socio-economic characteristics of the area housing types, household incomes, condition of existing physical infrastructure and prevailing social-economic activities in a specific flood affected area (Chen et al. 2021; Tang et al. 2021).

2.3.2. GIS-based spatial flooding modelling

The GIS approach to shoreline (coastal) flood risk assessment is a spatial analysis technique that is primarily based on topography. The so-called 'bathtub' approach represents hydraulic connectivity to the open lake and models the flooded area by selecting elevation values of a digital elevation model (DEM) which have a lower value than the specified maximum water level in the lake (Hadipour, Vafaie, and Deilami 2020; Sande Van de, Lansen, and Hoyng 2012; Seenath, Wilson, and Miller 2016).

The singular use of topographical data (i.e. a DEM) implies that other factors such as flow velocity and direction, bed friction, presence or absence of structural barriers, which are normally considered in physically-based hydrodynamic models, are not taken into account (Seenath, Wilson, and Miller 2016). Figure 4 presents a graphical illustration of the adopted flood modelling approach.

It assumes that if the water level rise approaches zones A from zone C, then zones A and C will be flooded even through zone B which separates C from A is higher and could potentially act as a flood barrier for Zone A. To mitigate this limitation, cells with hydraulic connectivity to the lake shoreline were selected and allowed to 'flood'; all cells that are not connected to the shoreline were considered 'unflooded' and removed.

Despite this limitation, the GIS approach to coastal flood modelling has been applied successfully in various flood risk assessment studies covering a large spatial extent with considerable levels of accuracy (Cohen et al. 2019; Sande Van de, Lansen, and Hoyng 2012). Furthermore, to improve the accuracy of the flood model results, the Forest and Buildings removed Copernicus Digital Elevation Model (FABDEM), a recent global DEM in which vegetation (such as trees) and buildings are filtered out, was used (Fathom 2021; Hawker et al. 2022).

Table 1. Details of various types of data used for developing the GIS based flood model.

Type of Data	Description	Source	Purpose
Observed daily water levels (m)	Daily record (water level 1900–2021)	Ministry of Water and Environment; Directorate of of Water Resources Management, Water Level data recorded at Jinja Pier Water Level Gauge	Statistical analysis and Input for flood modelling and mapping
Digital elevation model (FABDEM)	30m x 30m, Publication date: 17 December 2021. 0o − 10o N, 30o − 40o E	University of Bristol https://data.bris.ac.uk/data/dataset/ (accessed on 26 November 2022)	Elevation data for identifying the areas prone to flooding
Uganda shapefiles	Uganda shapefiles: districts, municipalities, parishes.	Uganda Bureau of Statistics (UBOS) Website: https://ubos.maps.arcgis.com/	Study area delineation, DEM extraction, identifying areas of interest by names.
Satellite image	ArcGIS Basemap "Imagery with Labels" 7 July 2021, in Raster format	Base map in ArcGIS	Digitization of the flood affected infrastructure
Google earth image	ArcGIS Basemap "OpenStreet Map"	Base map in ArcGIS	Location of the study area

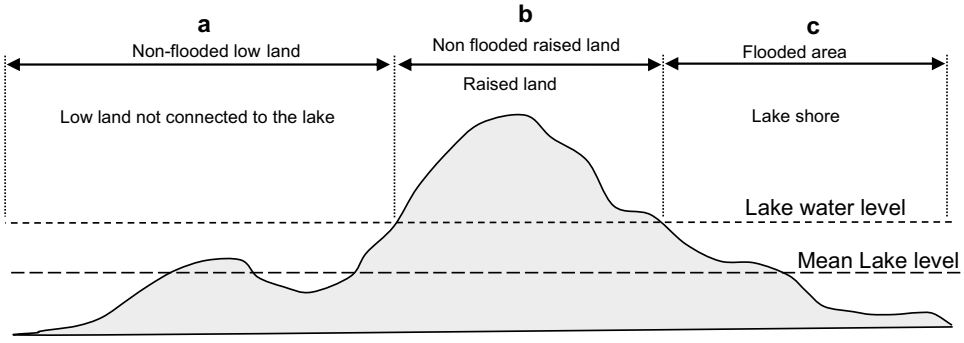


Figure 4. GIS based flood modelling approach. Adapted from Seenath et al., (2016).

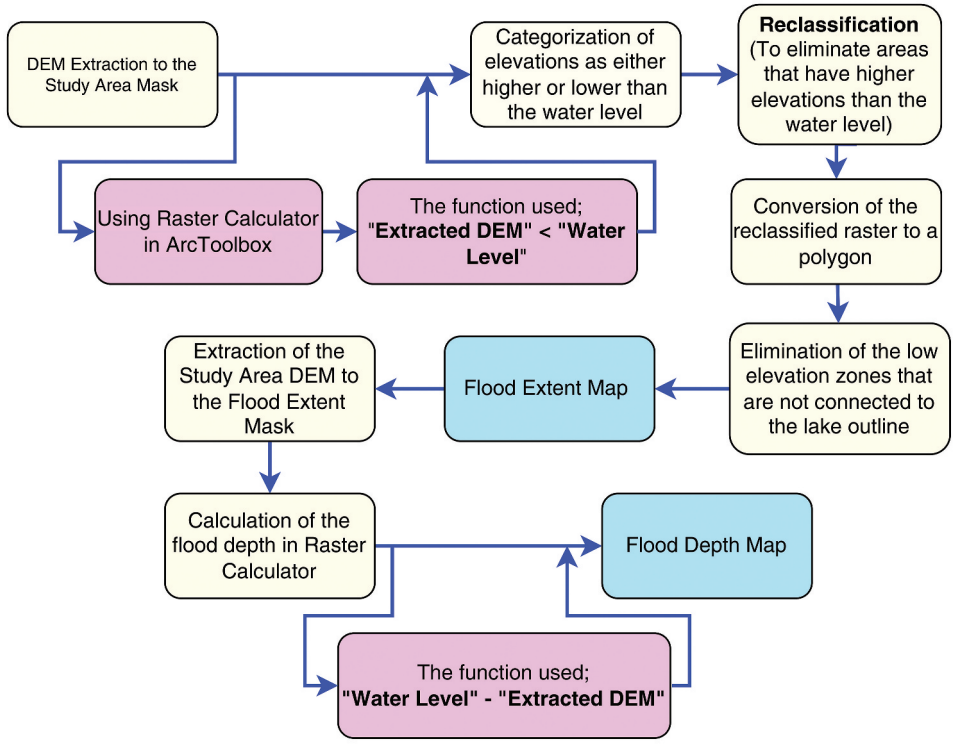


Figure 5. Schematic diagram of the GIS based spatial flood modelling approach.

The main steps entailed in the proposed GIS-based spatial flood modelling approach are illustrated in Figure 5. A GIS shapefile for Entebbe City was extracted from the Uganda Bureau of Statistics, (UBOS) GIS shapefile prepared for the whole of Uganda (HDX 2019). The extracted shape file was used as a boundary to clip the parishes within Entebbe City from the Uganda parishes shape file using 'clip' geoprocessing tool in ArcGIS. The resulting parish shapefile formed the boundary map of the study area and was used to clip the DEM for the study area. The raster calculator tool in ArcGIS toolbox was

used to identify cells on the DEM whose elevation was below the elevation of a specified lake water level rise. The expression used in the raster calculator was 'DEM' less than or equal to 'water surface level'. This identified all the cells on the DEM whose elevations were less than the considered water surface elevation. The identified patches refer to the area that can be flooded under a given lake water level rise scenario. The results were then exported as a flood extent map.

The flood depths resulting from a specified lake water level rise was determined as the difference between the considered water surface level and the DEM elevation at a given point. Specifically, the generated flood extent file was used to clip the DEM in order to generate the DEM relevant to the flooded area only. The elevations in the clipped DEM of the flooded area were subtracted from the surface water elevations to generate the flood depth map. In this research flood extent and depth maps were generated based on the projected Gumbel AMWL of 13.49 m ($T = 44$ yrs), which corresponds to the maximum water level observed in May 2021.

2.3.3. Flood consequence assessment

Figure 6 illustrates the adopted flooding consequence assessment approach. In this approach, different shape files were created in ArcGIS for the different types of buildings or infrastructure i.e. commercial, residential, institutional and industrial buildings and access roads. The flood extent and depth maps were overlain over a 2021 satellite image for the study area. Each type of building or infrastructure within the flooded area was digitized and the number of each type computed. Furthermore, the proportion of the population residing in the flood affected parishes was determined.

Lastly but not least, direct tangible flood damages were assessed qualitatively using questionnaire-based interviews of property owners in the flood affected parishes in Entebbe City. The direct tangible flood damage includes damage to property, e.g. floors, walls, ceilings, internal plumbing and electrical items among others, but do not include intangible flood damages such as loss of production and profit, disruption to road traffic and other flood related clean-up costs (Butler and Davies 2011).

2.4. Validation of simulated flooding impacts and consequences

2.4.1. Field surveys

In this study, a qualitative approach was applied to validate the resulting flooding impacts and consequences. Qualitative approaches, such as use of focus group discussions or individual interviews, have been successfully applied in validation of flood modelling outputs because they facilitate understanding of underlying reasons, opinions and motivations and therefore provide useful insights into past flooding event characteristics (Chen et al. 2021). In this study, individual interviews of selected property owners in the case study area were undertaken to obtain critical information of the resulting flooding consequences.

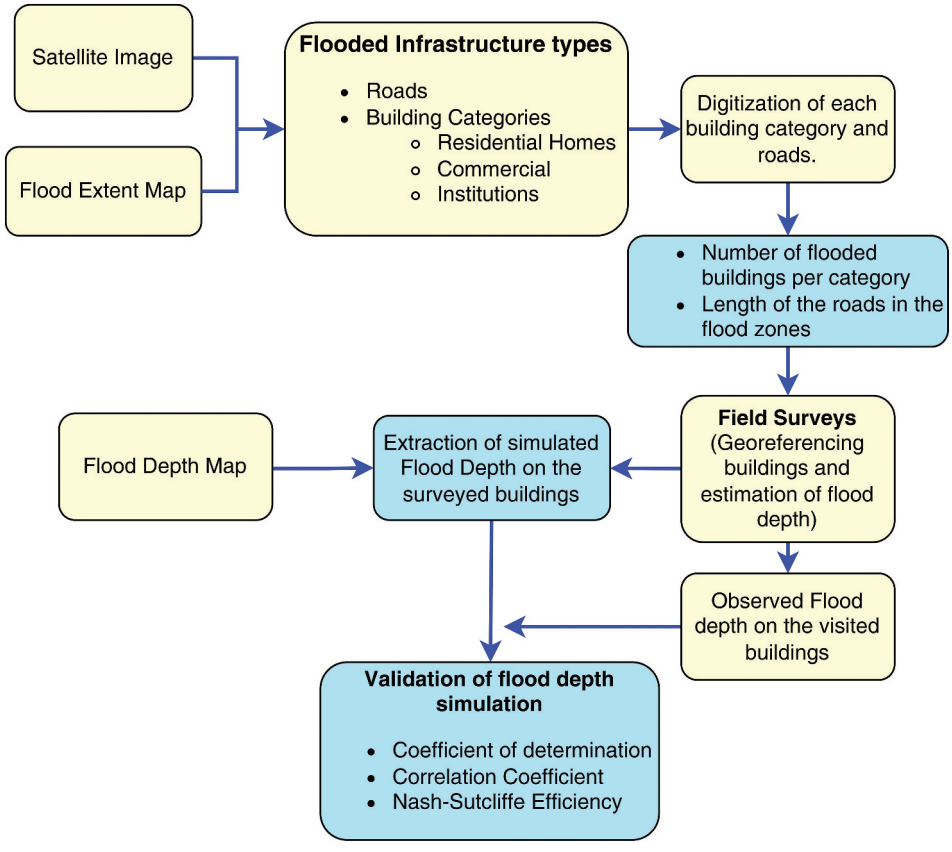


Figure 6. Schematic diagram of the flood consequence assessment approach.

The sample size was determined using Equation 67 (Arkin and Colton 1963)

$$\text{Sample size}, n = \frac{Z^2 N P_v (P_v - 1)}{M^2 (N-1) + Z^2 P_v (P_v - 1)} \quad (7)$$

with 95% confidence level $Z = 1.96$, degree of variability (P_v) = (50%), and M = level of precision or sampling error which is ±10%. Considering the population of 69,430 people in Entebbe (UBOS 2016), a sample size of 100 was computed.

Based on the computed sample size, field surveys which comprised of questionnaire surveys in which selected property owners in flood affected parishes were interviewed, and measurements of observed maximum flood depths based on identified high water (flood) marks and georeferencing of selected flood affected properties. Such an approach was applied in a recent study in which post-storm surveys on reported flooding impacts were undertaken and used to validate the flood modelling results (Perini et al. 2016).

The flooding impacts and consequences were assessed using questions that were focused on ascertaining the number and type of buildings and infrastructure that were affected, flood depth and direct damage to buildings. A total of 100 questionnaire-based interviews were undertaken.

2.4.2. Validation of simulated flooding impacts

Field measurements of the observed high-water marks at selected existing buildings in the study area were undertaken to obtain information on observed maximum flood depths. The maximum flood depth was determined as the distance between existing ground level and the highest water mark on a given flood affected building. In instances where maximum flood levels could not be easily ascertained, the property owners were requested to provide information on the observed maximum flood depths. Some property owners were able to specify the maximum flood depth through referencing of key existing building levels such as the skirting level, below window level or above the window level. Based on this, the maximum observed maximum flood depths were measured and recorded.

Furthermore, the coordinates of each surveyed building were obtained using a mobile phone GPS application. A total of 25 flood affected properties were georeferenced and the observed maximum flood depth measured. The measured maximum flood depth was compared to the modeled flood depth at each of the 25 selected buildings. The corresponding simulated flood depth at each of the buildings was extracted from the flood depth map and results compared. The Nash–Sutcliffe Efficiency (NSE), coefficient of determination (R^2) and correlation coefficient parameters were applied to assess the goodness of fit between the observed and simulated flood depths.

3. Results

3.1. Analysis of annual maximum lake water levels

The time series of the AMWLs is shown in Figure 7. The highest AMWLs occurred in 1964 (13.34 m) and May 2021 (13.49 m). On the other hand, the lowest annual maximum water level occurred in 1922 (10.5 m) and 2006 (10.86 m) respectively. The lake water levels have been rising since 2006 to the maximum observed level of 13.49 m in 2021, representing an average increase in lake water levels of 15.4% over a relatively short 3 year period, that is an average annual rise in lake water levels of 554 mm/yr.

3.2. Prediction of average recurrence interval of extreme AMWLs

3.2.1. Statistical modelling results

The R software, which is an open source statistical modelling software (de Micheaux, Pierre, and Liquet 2013) was used to determine the parameters for the Gumbel and GEV distributions (Table 2).

Table 2. Distribution parameters for the GEV and Gumbel distributions.

Distribution	Location Parameter	Scale parameter	Shape parameter
Gumbel Distribution	11.3334	0.5731	-
GEV distribution	11.3424	0.5800	−0.0292

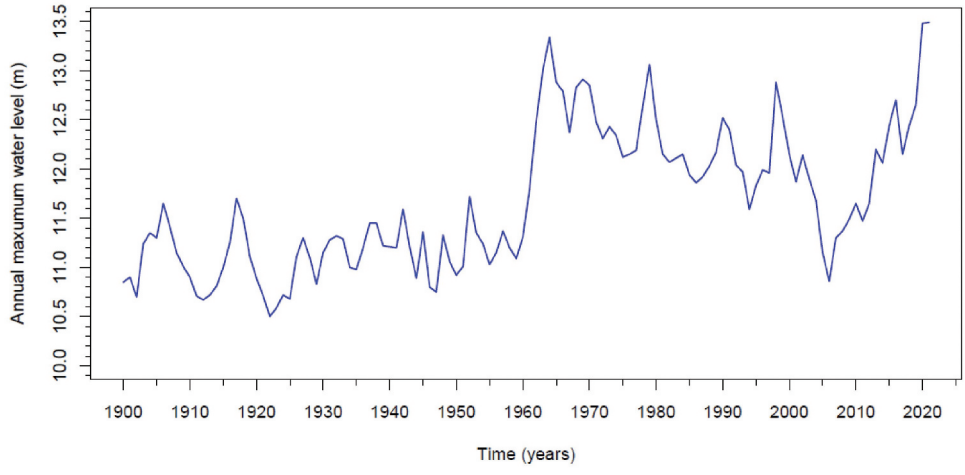


Figure 7. Variation of Lake Victoria water levels from 1900 to 2021.

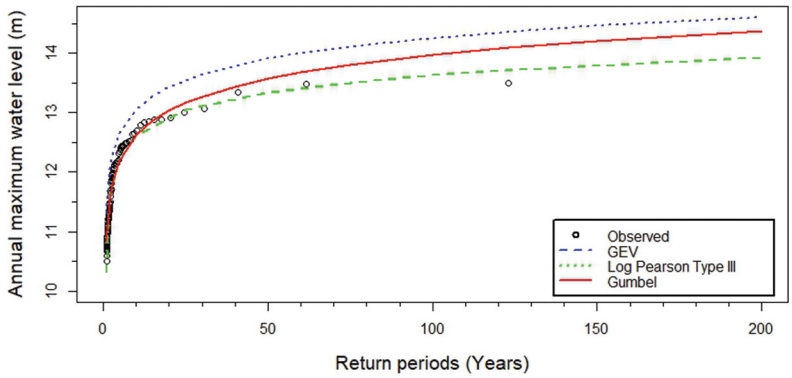


Figure 8. Comparison of predicted and observed AMWLs.

Table 3. Comparison of annual maximum water levels predicted for various return periods using Gumbel, GEV and log pearson type III statistical models.

	Computed lake water level return period in years						
Statistical Distribution	5	10	25	50	75	100	200
Gumbel (m)	12.19	12.62	13.17	13.57	13.80	13.97	14.37
GEV (m)	12.65	13.05	13.55	13.91	14.10	14.26	14.60
Log-Pearson Type III (m)	12.23	12.61	13.04	13.35	13.50	13.64	13.92
Annual Maximum Series (AMS) of observed water levels (m)	12.35	12.68	13.01	13.40	13.49	13.49	

The observed and predicted values of the AMWLs corresponding to various return periods are presented in Figure 8.

The Gumbel distribution, predicted the return period of the maximum observed water level of 13.49 m as 43.8 years (approximately 44 years). On the other hand, the Log-Pearson Type III and GEV distributions predicted the return period of the observed maximum water level of 13.49 m as 75 years and 27.5 years respectively.

Table 3 presents the computed return periods corresponding to various maximum water levels for the three considered statistical models. From the table, it can be noted that the GEV generally underestimated the return periods of the AMWLs. On the other hand, the Log Pearson Type III distribution tended to overestimate the corresponding return period of the observed AMWLs.

3.2.2. Assessment of goodness of fit of the statistical models

Table 4 presents the computed values of the Nash-Sutcliffe Efficiency index (NSE), the root mean square error (RMSE) and the correlation coefficients that were computed for the Gumbel, GEV and Log-Pearson Type III statistical distributions.

Table 4. Computed Nash-Sutcliffe Efficiency index (NSE), root mean square error and correlation coefficients for Gumbel, GEV and Log-Pearson type III statistical models.

	Statistical distribution model		
Performance criteria	Gumbel	GEV	Log-Pearson Type III
Nash–Sutcliffe Efficiency Index (NSE)	0.975	0.559	0.883
Root Mean Square Error (RMSE)	0.111	0.469	0.242
Correlation Coefficient	0.988	0.990	0.989

Based on the NSE parameters, the Gumbel statistical distribution with an NSE value of 0.975 provides the best fit of the observed AMWLs, followed by the Log Pearson Type III distribution (NSE = 0.883) and GEV distribution (NSE = 0.558).

The Gumbel distribution resulted into the lowest RMSE value of 0.111, followed by the Log-Person Type III distribution with a RMSE value of 0.242. On the other hand, the GEV distribution had a higher RMSE value of 0.469 implying that the distribution did not provide a good fit for the observed AMWLs. However, the computed correlation coefficients for the Gumbel, GEV and Log-Pearson Type III distributions were 0.988, 0.990, 0.989, respectively, which were within a close range.

Based on the results above, it can be concluded that, of the three statistical models that were applied in the study, the Gumbel Distribution provided the best fit to the observed AMWLs for Lake Victoria.

Furthermore, based on the results of the Gumbel distribution, it can be concluded the maximum water level (MWL) that was observed in May 2021 has a return period, T of approximately 44 years.

3.3. GIS based spatial flood modelling results

3.3.1. Simulated flood extent

The GIS based spatial flood modelling techniques described in Section 2 were applied to simulate the flooding extent (flooded area) for Entebbe City based on the $T = 44$ yr Gumbel AMWL of 13.49 m (Figure 9). The total flooded area was determined as 9.1 km^2 which amounts to 25% of the total area in Entebbe City (Table 5). The results also indicate that the GEV distribution leads to a considerable over estimation of the flooded area (10%), while the Log-Pearson Type III slightly underestimates the flood depths by 9% (Table 6). It should however be noted that a significant proportion of this flooded land area mostly

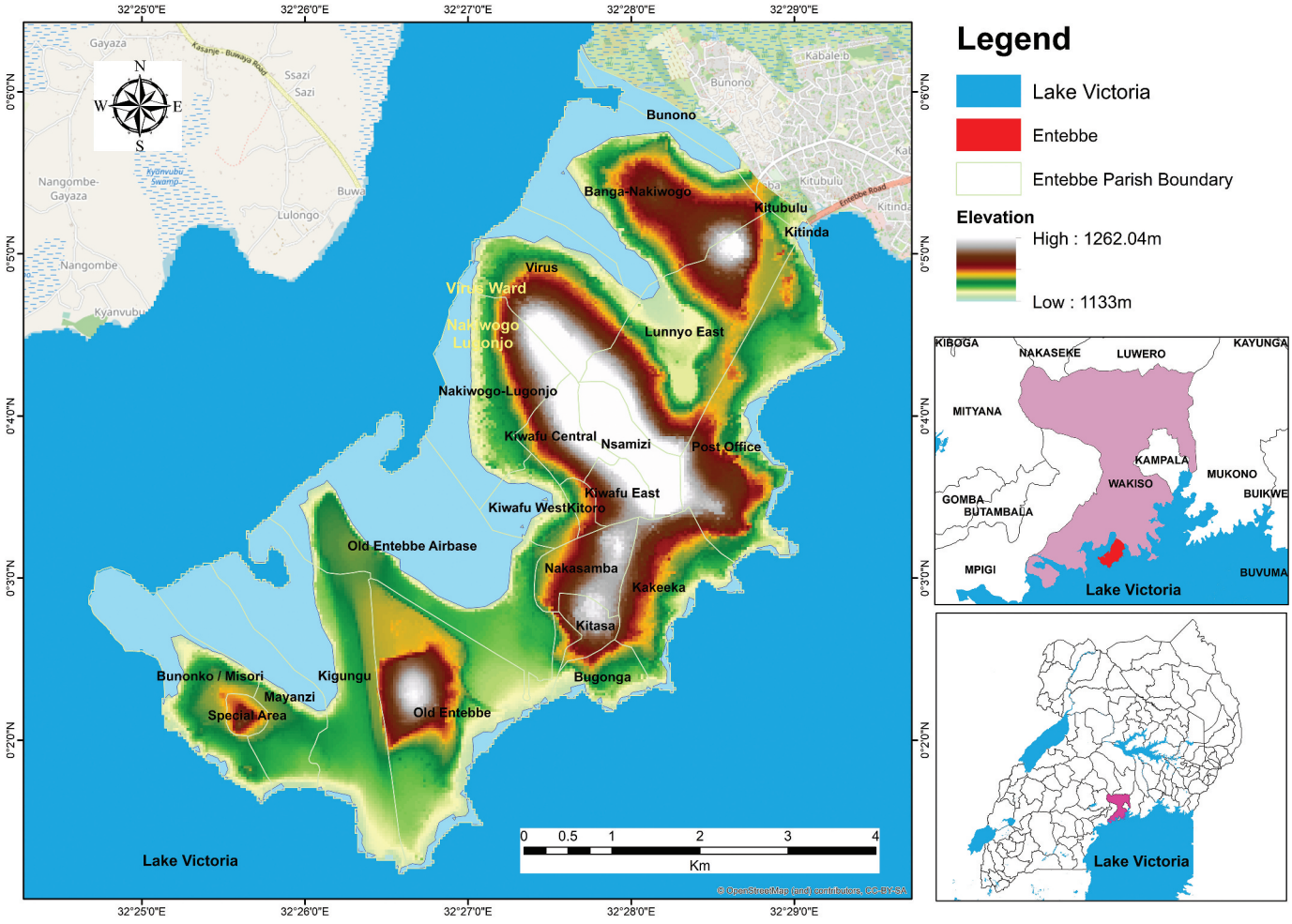


Figure 9. Flood extent map considering Gumbel $T = 44$ yr AMWLs.

Table 5. Simulated flooding extent in various parishes in Entebbe City.

Parish	Parish area (km^2)	Flooded area (km^2)	% Flooded area
Kitubulu	0.128	0.006	4.69
Bunono	0.462	0.360	77.87
Kitinda	0.013	0.001	8.62
Post Office	2.913	0.180	6.18
Bugonga	0.539	0.027	5.01
Kitasa	0.335	0.000	0.00
Nakasamba	0.801	0.000	0.00
Old Entebbe	3.444	0.860	24.97
Old Entebbe Airbase	4.417	2.520	57.05
Kigungu	3.226	1.170	36.27
Mayanzi	2.062	0.970	47.04
Special Area	0.185	0.000	0.00
Kitoro	0.167	0.000	0.00
Kiwafu East	0.427	0.000	0.00
Kiwafu Central	0.672	0.000	0.00
Kiwafu West	0.726	0.270	37.21
Nakiwogo-Lugonjo	2.324	0.800	34.43
Kakeeka	1.450	0.059	4.07
Nsamizi	0.760	0.000	0.00
Lunnyo East	2.999	0.003	0.10
Virus	2.089	0.190	9.10
Banga-Nakiwogo	3.544	1.040	29.34
Bunonko/Misori	1.898	0.640	33.72
Total	**35.58**	**9.10**	**25.56**

comprises of wetlands located on the north-western side of the Entebbe peninsular (Magumba, Takagaki, and Kikuchi 2014). This suggests that wetland protection should be prioritised as a suitable flood resilience enhancement strategy in Entebbe City.

3.3.2. Simulated flood depths

Figure 10 presents the results of the GIS based spatial modelling of flood depths. The model results based o the Gumbel distribution, and resulted in average flood depths of 1.52 [0.5–3.5] m. The results also indicate that the GEV distribution leads to a significant over estimation of the resulting flood depths (38%), while the Log-Pearson Type III slightly underestimates the flood depths by 4% (Table 6). The results also indicate that the highest flood depths occurred on the north-western shoreline of Entebbe which has large tracts of wetlands and low-lying flood prone areas compared to the south eastern shoreline with a greater length of cliffs and steep slopes.

3.3.3. Assessment of flooding consequences

\Based on the simulated flooding extent and depths, digitization of properties and infrastructure in the flooded area was undertaken by overlaying the GIS shape file of the flooding extent over a high-resolution satellite image for Entebbe City (Figure 11). Using this approach, the total number of affected buildings and infrastructure that were affected by the flooding were determined as follows: 40 commercial buildings, 204 residential buildings and 23 markets.

In addition, an estimated 8.9 km of urban roads became impassable on account of the flood waters. Furthermore, it was also reported that over 2000 residents in Nakiwogo

Table 6. Simulated flooded areas and depths resulting from the projected 44-year annual maximum lake level for Gumbel, GEV and Log-Pearson type III statistical distributions.

Statistic Distribution	44-Year annual max. water level (m)	Flooded area (km^2)	Max. Flood depth (m)
Gumbel	13.49	9.10	3.50
GEV	13.83	10.00	4.83
Log-Pearson Type III	13.35	8.29	3.35

parish were affected by the flood waters (The Independent 2020). Current National Environmental Management Authority (NEMA) guidelines state that a buffer zone of 200 m (where no development should take place) should be maintained on lake shorelines. In Figure 12 the purple line represents the extent of the proposed buffer zone which was determined through offset of a horizontal distance of 200 m from the boundary of flood extent lines.

3.4. Validation results

3.4.1. Field survey findings

The qualitative survey results indicated that over 50% of the interviewed people suffered significant direct flood damage, with an average direct flood damage of US$ 108.40 per household interviewed. The survey results also indicated that the floods affected mainly residential and commercial properties. Furthermore, 50% of the residents whose residential properties were damaged reported an average flood duration ranging from 1–2 years which suggests that the persistent nature of floods caused by the lake water level rise.

The average flood depth which was measured based on the observed high-water marks on selected buildings was 1.16 [0.3–2.1] m above the floor level (24% of the surveyed households recorded the highest watermark below the skirting, 64% of homes recorded the highest watermark between the skirting and the window sill level while the remaining 12% was above the window sill level).

The major flooding impacts on buildings and infrastructure that were reported included settlement of buildings, total collapse of buildings, ingress of water in building electrical wiring, and corrosion of steel members of structures such as doors and windows.

Lastly but not least, approximately 10% of respondents reported flood related injuries and an increase in water-related diseases such as malaria, skin diseases, allergies and trench foot.

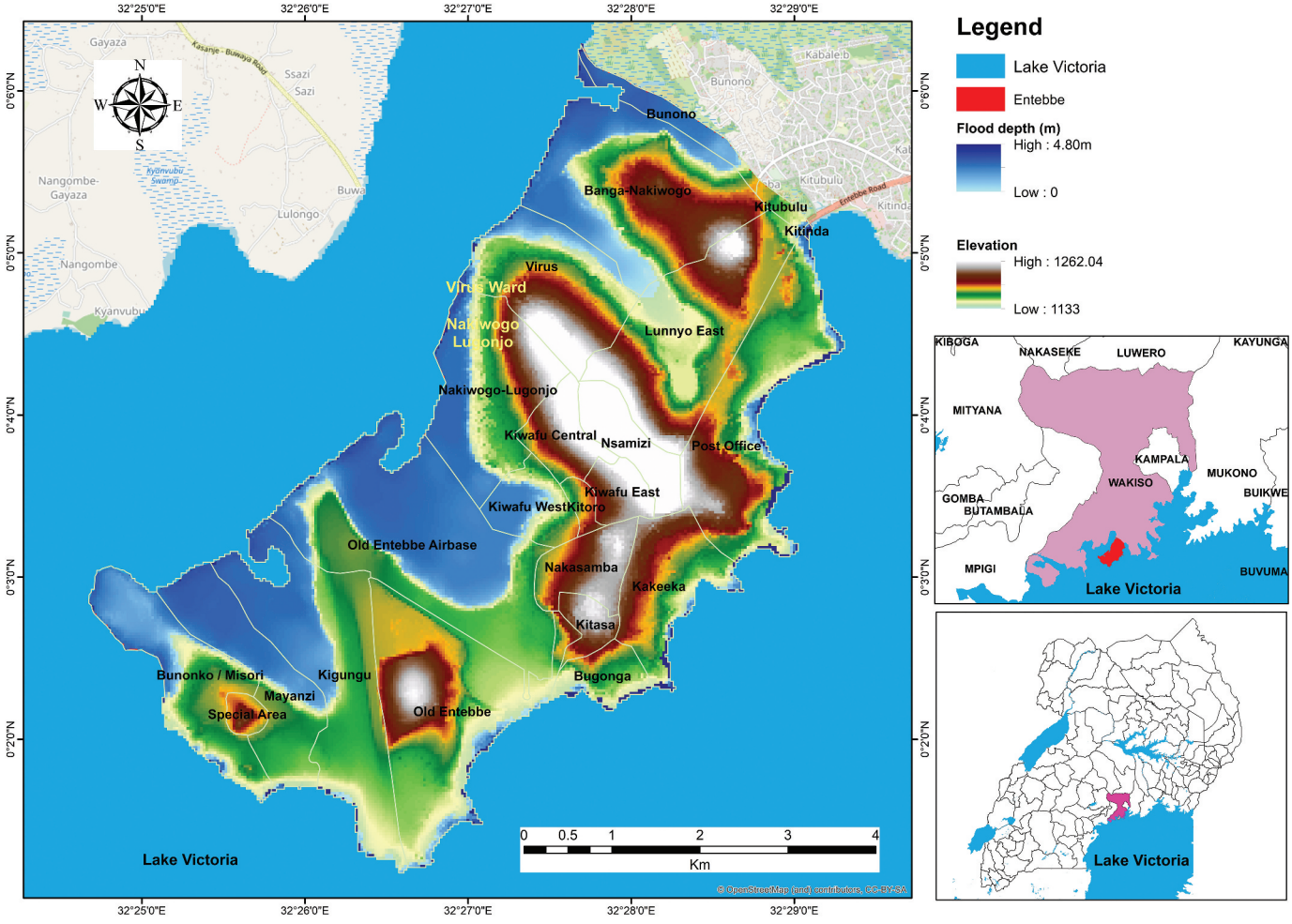


Figure 10. Simulated flood depths considering Gumbel $T = 44$ yr AMWLs.

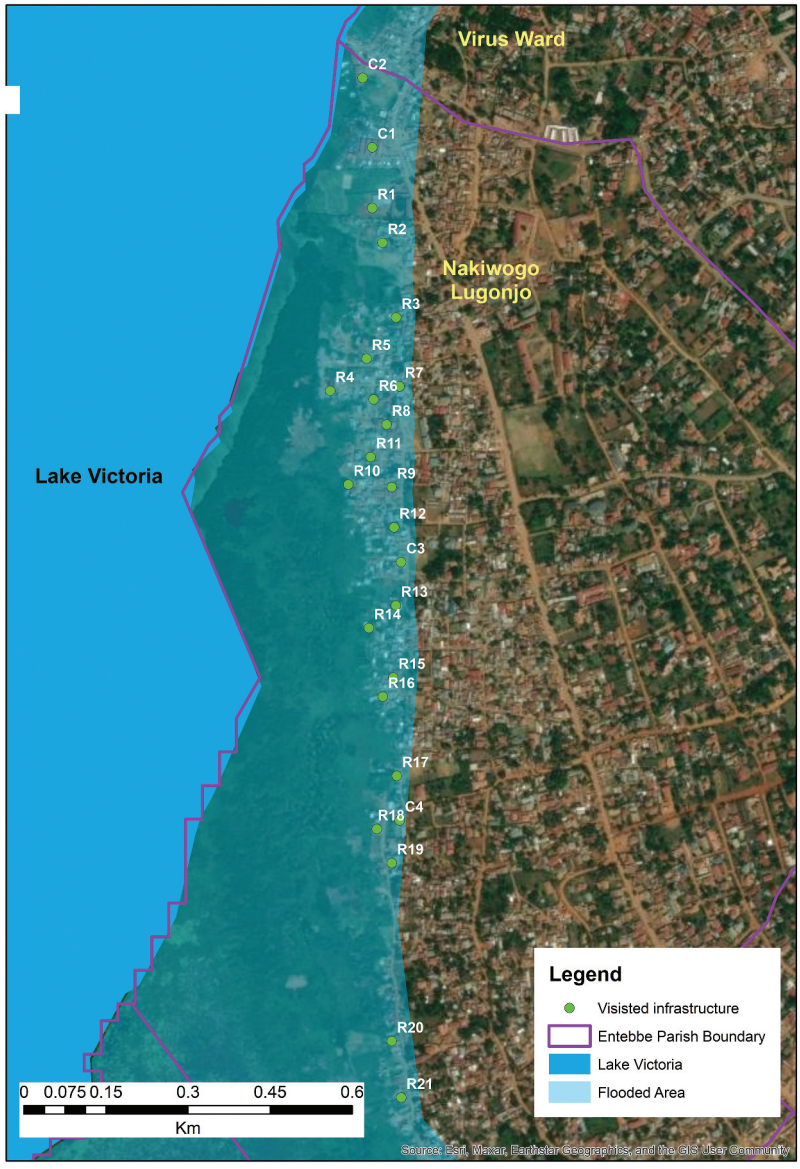


Figure 11. Proposed flood buffer zone.

3.5. Validation of simulated flooding impacts

Table 7 presents the observed flood depths at selected buildings/infrastructure and the corresponding simulated flood depths. The results indicate that the simulated flood depths are higher than the observed flood depth by approximately 31%. The scatter plot in Figure 13 shows a good correlation between the modeled and observed flood depth. Furthermore, the NSE, coefficient of determination and the correlation coefficient were computed to assess goodness of fit between the simulated and the observed flood depths. The results for the NSE, coefficient of determination and the correlation coefficient were 0.56, 0.91 and 0.95, respectively.

4. Results discussion

The results of the lake water level analysis confirmed that rapid rise of 15.4% (1.8 m) in lake water levels occurred over a relatively short period of 3.2 years and represented an annual rise of 554 mm/yr. The results of the statistical modelling confirmed that the Gumbel distribution provided the best fit of the observed AMWLs. Using the Gumbel distribution, the return period of observed maximum lake water level rise was determined as 44 years. However, previous studies have suggested decadal (11-year) cyclic variations in Lake Victoria water levels which were attributed to changes in rainfall patterns driven by strong sun spot–rainfall relationships and El Nino that results from the seasonal passage of the Intertropical Convergence Zone (ITCZ) (Stager et al. 2007). Based on the 11-year cycles, and the minimum lake water level recorded in 2006, the subsequent peak lake water levels would be expected in 2017 and 2028 respectively. This therefore suggests that the lake water level observed in May 2021 was not only caused by natural climate variability but could also be linked to the emerging effects of climate change that have led to an increase in the frequency and magnitude of extreme rainfall events in the Lake Victoria basin. The rising lake water levels could also be attributed to sub-optimal regulation of lake water releases at Jinja. This therefore suggests the need for further research focused on optimization of lake water releases and revision of the 'agreed curve' as a complementary strategy for maintaining near normal lake water levels and minimization of attendant flooding impacts and consequences.

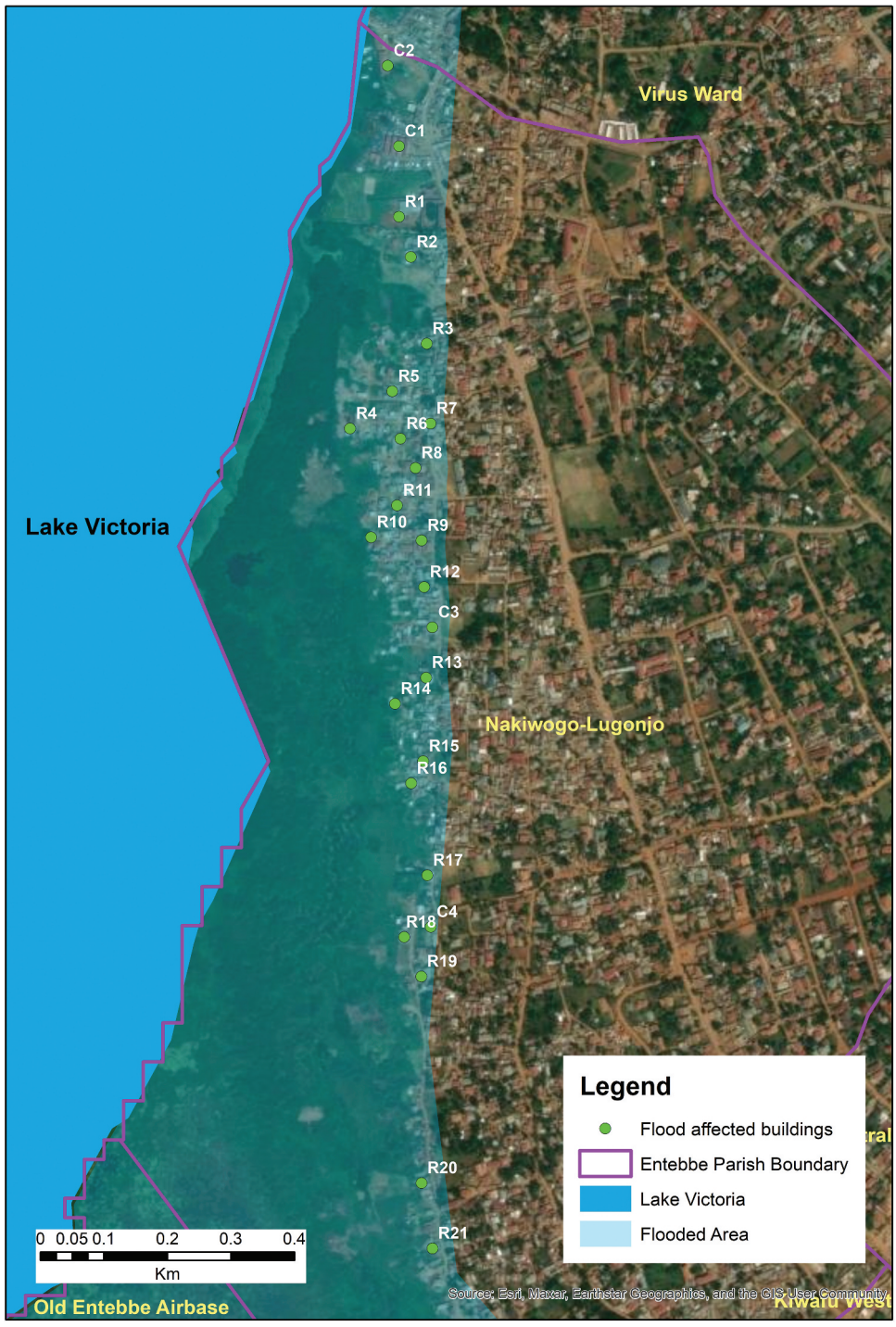


Figure 12. Resulting flood extent map showing the flood affected buildings and infrastructure.

Based on the results of the GIS-based spatial flood modelling, the total flooded area and average flood depth were determined as 9.1 km^2 (representing a total flooded area of 25%) and 1.52 m respectively. The flooding impacts were mostly concentrated on the north-western shoreline of Entebbe which has large tracts of low-lying flood prone areas. The results therefore suggest that topographical factors, such as elevation and slope of land adjacent to the lake, significantly influences the risk of lake water level induced flooding. For example, the same lake water level rise will not significantly affect a lake shoreline with high cliffs when compared to one with very flat and low-lying landscape (e.g. Chen et al. 2021). The field surveys confirmed the flooding lasted between 1–2 years which indicates that ake water level induced flooding is very persistent unlike urban pluvial flooding whose corresponding average flood duration is in hours. Furthermore, the results also suggest that the presence of natural wetlands adjacent to the lake shore also provides an additional buffer storage capacity, that minimizes the resulting flooding consequences. This therefore confirms the need to focus on

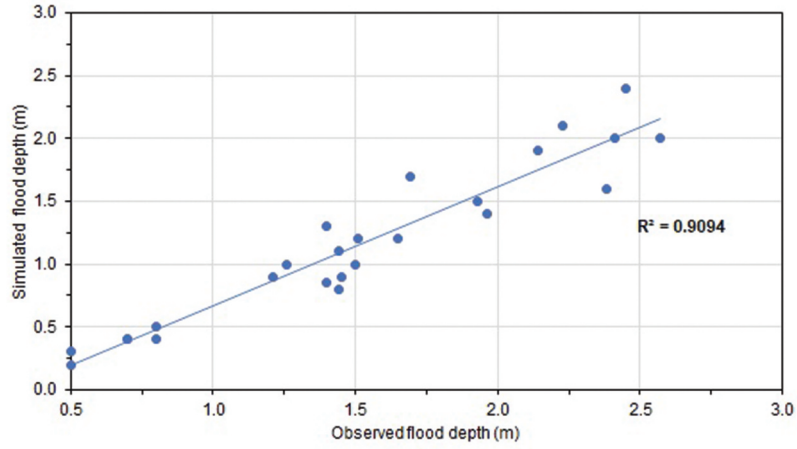


Figure 13. Scatter plot of observed versus simulated flood depths.

Table 7. Observed and simulated flood depths for selected buildings in Nakiwogo parish.

ID	Building Type	X Coordinate	Y Coordinate	Simulated flood depth (m)	Observed flood depth (m)
C1	Commercial	438752.80	438752.80	2.57	2.00
R1	Residential	438752.80	438752.80	1.26	1.00
R2	Residential	438771.33	438771.33	1.21	0.90
C2	Commercial	438735.08	438735.08	2.45	2.40
R3	Residential	438796.46	438796.46	0.80	0.40
R4	Residential	438675.68	438675.68	1.50	1.00
R5	Residential	438742.35	438742.35	1.45	0.90
R6	Residential	438755.05	438755.05	1.40	0.85
R7	Residential	438802.68	438802.68	0.50	0.20
R8	Residential	438778.87	438778.87	0.70	0.40
R9	Residential	438788.39	438788.39	0.50	0.30
R10	Residential	438709.02	438709.02	1.93	1.50
R11	Residential	438749.76	438749.76	1.44	1.10
R12	Residential	438792.62	438792.62	0.80	0.50
C3	Commercial	438805.32	438805.32	1.44	0.80
R13	Residential	438795.80	438795.80	1.65	1.20
R14	Residential	438746.59	438746.59	2.41	2.00
R15	Residential	438791.04	438791.04	1.96	1.40
R16	Residential	438771.99	438771.99	1.69	1.70
C4	Commercial	438802.81	438802.81	1.40	1.30
R17	Residential	438797.52	438797.52	0.70	0.40
R18	Residential	438761.14	438761.14	2.38	1.60
R19	Residential	438788.26	438788.26	1.51	1.20
R20	Residential	438788.32	438788.32	2.23	2.10
R21	Residential	438805.79	438805.79	2.14	1.90
Average Flood Depth (m)				**1.52**	**1.16**

adaptation strategies that combine wetland conservation and restoration with the traditional flood defense structures such as dikes and flood walls (Temmerman et al. 2023).

Lastly but not least, the results of the model validation suggested that although the adopted methodology resulted into slightly higher flood depths, the GIS-based spatial flood modelling approach still provides a robust and reliable approach for assessment of lake water level induced flooding for inland lakes, if coupled with detailed field surveys and data collection for model validation.

5. Conclusions and recommendations

This research has developed and applied a combination of statistical modelling and GIS-based spatial flood modelling approaches to quantify urban flooding impacts and consequences in Entebbe city, Uganda, caused by the recent rise in Lake Victoria water levels. Using the validated GIS-based spatial flood model, the resulting flooding extent and average flood depths were determined. In addition, through application of GIS overlay analysis, the total number of affected buildings and length of urban roads were quantified. The proposed methodology provides a robust and computationally efficient approach for identification of at-risk buildings along the lake shoreline and for testing effectiveness of potential adaptation strategies.

Furthermore, although the approach does not utilise physical-based modelling techniques that consider hydrodynamic factors such as flow velocity and direction, bed friction and presence or absence of structural barriers, it still provides a robust and computationally efficient approach that can be used to identify at-risk buildings along inland lake shorelines, given the availability

of high resolution digital elevation models, recent satellite imagery and post flood field survey data for the study area.

Last, but not least, the developed coastal flood modelling approach and results can provide guidance to authorities such as the Nile Basin Initiative, the Lake Victoria Basin Commission and national level Environmental Management Authorities in the three east African countries to develop policies for sustainable management of the lake water resource in view of emerging climate change and urbanisation threats. This could be operationalised through the delineation of flood buffer zones for building control purposes, relocation of flood affected persons, and restriction of new developments in the delineated flood buffer zones. The study outputs could also be used to facilitate grass root stakeholder engagements focused on promotion of community level flood resilience measures such adoption of property flood resilience measures and implementation of wetland protection and restoration measures as suitable nature based solutions (CIRIA 2021; Ferreira et al. 2022; Temmerman et al. 2023)

Future research focused on (a) comparison of the proposed approaches with physically-based coastal hydrodynamic models that consider water wave motion, flow velocities and surface roughness and (b) testing effectiveness of potential adaptation strategies such as construction of flood walls or nature-based solutions such as constructed wetlands in flood buffer zones is required.

Acknowledgements

Acknowledgement is given to the Directorate of Water Resources Management (DWRM), Ministry of Water and Environment, Uganda and Uganda Electricity Generation Company Ltd (UEGCL) for provision of the observed water level data for Lake Victoria. The insights of the three anonymous reviewers are also gratefully acknowledged. We also confirm that there are no relevant financial or non-financial competing interests to report.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Seith N. Mugume http://orcid.org/0000-0002-8289-0099
David Butler http://orcid.org/0000-0001-5515-3416

References

Arkin, H., and R. R. Colton. 1963. *Tables for Statisticians*. 2nd Edition ed. New York: Barnes and Noblebooks.

Butler, David, and J. W. Davies. 2011. *Urban Drainage*. 3rd ed. London and New York: Spon Press, Taylor and Francis Group.

Chen, Yun, Rui Liu, Damian Barrett, Lei Gao, Mingwei Zhou, Luigi Renzullo, and Irina Emelyanova. 2021. "A Spatial Assessment Framework for Evaluating Flood Risk Under Extreme Climates." *Science of the Total Environment, The* 538:512–523. https://doi.org/10.1016/j.scitotenv.2015.08.094.

CIRIA. 2021. *Code of Practice for Property Flood Resilience C790*. London, UK: Construction Industry Research and Information Association.

Cohen, Sagy, Austin Raney, J. Derek Loftis Dinuke Munasinghe, Andrew Molthan, Jordan Bell, Laura Rogers, et al. 2019. "The Floodwater Depth Estimation Tool (FwDet V2.0) for Improved Remote Sensing Analysis of Coastal Flooding." *Natural Hazards and Earth System Sciences* 19 (9): 2053–2065. https://doi.org/10.5194/nhess-19-2053-2019.

Cunnane, C. 1973. "A Particular Comparison of the Annual Maxima and Partial Durations Series Methods of Flood Frequency Prediction." *Hydrology* 18 (3–4): 257–271. https://doi.org/10.1016/0022-1694(73)90051-6.

Dawson, R. J., L. Speight, J. W. Hall, S. Djordjević, D. A. Savić, and J. Leandro. 2008. "Attribution of Flood Risk in Urban Areas." *Journal of Hydroinformatics* 10 (4): 275–288. https://doi.org/10.2166/hydro.2008.054.

de Micheaux, Lafaye, Rémy Drouilhet Pierre, and Benoit Liquet. 2013. "The R Software: Fundamentals of Programming and Statistical Analysis." Statistics and Computing. http://link.springer.com/10.1007/978-1-4614-9020-3.

Fathom. 2021. "FABDEM a 30 M Global Map of Elevation with Forests and Buildings Removed." https://www.fathom.global/product/fabdem/ .

Feloni, Elissavet, Ioannis Mousadis, and Evangelos Baltas. 2020. "Flood Vulnerability Assessment Using a GIS-Based Multi-Criteria Approach—The Case of Attica Region." *Journal of Flood Risk Management* 13 (S1): 1–15. https://doi.org/10.1111/jfr3.12563.

Ferreira, Carla Sofia Santos, Kristina Potočki, Marijana Kapović-Solomun, and Zahra Kalantari. 2022. "Nature-Based Solutions for Flood Mitigation and Resilience in Urban Areas." In *Handbook of Environmental Chemistry*, Vol. 107, pp. 59–78. Springer Link. https://doi.org/10.1007/698_2021_758.

Funk, Chris, Pete Peterson, Martin Landsfeld, Diego Pedreros, James Verdin, Shraddhanand Shukla, Gregory Husak, et al. 2015. "The Climate Hazards Infrared Precipitation with Stations—A New Environmental Record for Monitoring Extremes." *Scientific Data* 2 (1): 1–21. https://doi.org/10.1038/sdata.2015.66.

Gayen, Sumita, Ismael Vallejo Villalta, and Sk Mafizul Haque. 2022. "Flood Risk Assessment and Its Mapping in Purba Medinipur District, West Bengal, India." *Water* 14 (1049): 1–19. https://doi.org/10.3390/w14071049.

Güneralp, Burak, Shuaib Lwasa, Hillary Masundire, Susan Parnell, and Karen C Seto. 2018. "Urbanization in Africa: Challenges and Opportunities for Conservation." *Environmental Research Letters* 13 (1): 015002. https://doi.org/10.1088/1748-9326/aa94fe.

Gupta, Hoshin Vijai, and Harald Kling. 2011. "On Typical Range, Sensitivity, and Normalization of Mean Squared Error and Nash-Sutcliffe Efficiency Type Metrics." *Water Resources Research* 47 (10): 2–4. https://doi.org/10.1029/2011WR010962.

Hadipour, Vahid, Freydoon Vafaie, and Kaveh Deilami. 2020. "Coastal Flooding Risk Assessment Using a GIS-Based Spatial Multi-Criteria Decision Analysis Approach." *Water* 12 (9): 2379. https://doi.org/10.3390/W12092379.

Hawker, L., P. Uhe, L. Paulo, J. Sosa, J. Savage, C. Sampson, and J. Neal. 2022. "A 30m Global Map of Elevation with Forests and Buildings Removed." *Environmental Research Letters* 17 (24016): 1–11. https://doi.org/10.1088/1748-9326/ac4d4f.

HDX. 2019. "Uganda Administrative Boundaries." https://data.humdata.org/dataset/uganda-administrative-boundaries-as-of-17-08-2018.

Herrnegger, Mathew, Gabriel Stecher, Christian Schwatke, and Luke Olang. 2021. "Hydroclimatic Analysis of Rising Water Levels in the Great Rift Valley Lakes of Kenya." *Journal of Hydrology: Regional Studies* 36 (May): 100857. https://doi.org/10.1016/j.ejrh.2021.100857.

Hirabayashi, Yukiko, Masahiro Tanoue, Orie Sasaki, Xudong Zhou, and Dai Yamazaki. 2021. "Global Exposure to Flooding from the New CMIP6 Climate Model Projections." *Scientific Reports* 11 (3740): 1–7. https://doi.org/10.1038/s41598-021-83279-w.

Hoffman, Zachary. 2021. *Land Use as a Predictor of Water Hyacinth (Eichhornia Crassipes) Presence on the Entebbe Coast of Lake*. Independent Study Project (ISP) Collection 3408. Brattleboro, Vermont, United States: School for International Training (SIT). https://digitalcollections.sit.edu/isp_collection/3408.

Huang, Yijing, Zhan Tian, Ke Qian, Junguo Liu, Masoud Irannezhad, Dongli Fan, Meifang Hou, and Laixiang Sun. 2020. "Nature-Based Solutions for Urban Pluvial Flood Risk Management." *Wiley Interdisciplinary Reviews: Water* 7 (3): 3. https://doi.org/10.1002/WAT2.1421.

Hwang, Seok Hwan, Dae Heon Ham, and Joong Hoon Kim. 2012. "A New Measure for Assessing the Efficiency of Hydrological Data-Driven Forecasting Models." *Hydrological Sciences Journal* 57 (7): 1257–1274. https://doi.org/10.1080/02626667.2012.710335.

The Independent. 2020. "Over 2000 Nakiwogo Residents Affected by Lake Victoria Floods." *Independent Newspaper*, April 27, 2020. https://www.

independent.co.ug/over-2000-nakiwogo-residents-affected-by-lake-victoria-floods/.

Inne, Vanderkelen, Nicole P.M Van Lipzig, and Wim Thiery. 2018a. "Modelling the Water Balance of Lake Victoria (East Africa) - Part 2: Future Projections." *Hydrology and Earth System Sciences* 22 (10): 5527–5549. https://doi.org/10.5194/hess-22-5527-2018.

Inne, Vanderkelen, Nicole P.M. Van Lipzig, and Wim Thiery. 2018b. "Modelling the Water Balance of Lake Victoria (East Africa)-Part 1: Observational Analysis." *Hydrology and Earth System Sciences* 22 (10): 5509–5525. https://doi.org/10.5194/hess-22-5509-2018.

Khaki, Mehdi, and Joseph Awange. 2021. "The 2019–2020 Rise in Lake Victoria Monitored from Space: Exploiting the State-Of-The-Art GRACE-FO and the Newly Released ERA-5 Reanalysis Products." *Sensors* 21 (13): 13. https://doi.org/10.3390/s21134304.

Kite, G. W. 1982. "Analysis of Lake Victoria Levels." *Hydrological Sciences Journal* 27 (2): 99–110. https://doi.org/10.1080/02626668209491093.

Kizza, Michael. 2012. *Uncertainty Assessment in Water Balance Modelling for Lake Victoria*. Makerere University, Kampala PhD Thesis.

Lanxin, Hu, Efthymios I Nikolopoulos, Francesco Marra, and Emmanouil N Anagnostou. 2020. "Sensitivity of Flood Frequency Analysis to Data Record, Statistical Model, and Parameter Estimation Methods: An Evaluation Over the Contiguous United States." *Journal of Flood Risk Management* 13 (1): 1–13. https://doi.org/10.1111/jfr3.12580.

Madsen, H., P. F. Rasmussen, and D. Rosbjerg. 1997. "Comparison of Annual Maximum Series and Partial Duration Series Methods for Modeling Extreme Hydrologic Events. 1. At-Site Modeling." *Water Resources Research* 33 (4): 747–757. https://doi.org/10.1029/96WR03848.

Mafaranga, Hope. 2020. "Heavy Rains, Human Activity, and Rising Waters at Lake Victoria." *Eos Magazine of AGU Advancing Earth and Space Science*. https://doi.org/10.1029/2020EO146582.

Magumba, David, Michiko Takagaki, and Masao Kikuchi. 2014. "Spatio-Temporal Changes in Wetlands and Identification of Cyperus Papyrus on the Northern Shore of Lake Victoria." *Tropical Agriculture and Development* 58 (1): 1–7.

Akurut, Mary, Patrick Willems, and Charles B. Niwagaba. 2014. "Potential Impacts of Climate Change on Precipitation Over Lake Victoria, East Africa, in the 21st Century." *Water* 6 (9): 2634–2659. https://doi.org/10.3390/w6092634.

Mugo, Robinson, Rose Waswa, James W. Nyaga, Antony Ndubi, Emily C. Adams, and Africa I. Flores-Anderson. 2020. "Quantifying Land Use Land Cover Changes in the Lake Victoria Basin Using Satellite Remote Sensing: The Trends and Drivers Between 1985 and 2014." *Remote Sensing* 12 (2829): 1–17. https://doi.org/10.3390/rs12172829.

MWE. 2020. *Statement on Rising Lake Victoria Water Levels*. Kampala: Uganda Media Centre. https://www.mediacentre.go.ug/media/statement-rising-water-levels-lake-victoria-and-nile-system.

NASA. 2020. *Lake Victoria Rising Water Levels (Washington, DC, USA: National Aeronautics and Space Administration)*.https://storymaps.arcgis.com/stories/bd820937c06845faa86f7f8944d56f47.

NASA. 2021. *Lake Victoria's Rising Waters*. National Aeronautics and Space Administration. https://earthobservatory.nasa.gov/images/148414/lake-victorias-rising-waters.

NBI. 2020. "Unprecedented Rise in Water Levels of Lake Victoria" In *NBI*, Entebbe, Uganda:Nile Basin Initiative Secretariat (Nile-SEC).Lake level rise in,higher than its average value. https://nilebasin.org/new-and-events/307-unprecedented-rise-in-water-levels-of-lake-victoria#:~:text=The.

Ntajal, Joshua, Benjamin L Lamptey, Ibrahim B Mahamadou, and Benjamin K Nyarko. 2017. "Flood Disaster Risk Mapping in the Lower Mono River Basin in Togo, West Africa." *International Journal of Disaster Risk Reduction* 23:93–103. https://doi.org/10.1016/j.ijdrr.2017.03.015.

Olaka, Lydia A., Joseph O. Ogutu, Mohammed Y. Said, and Christopher Oludhe. 2019. "Projected Climatic and Hydrologic Changes to Lake Victoria Basin Rivers Under Three RCP Emission Scenarios for 2015–2100 and Impacts on the Water Sector." *Water* 11 (7): 1449. https://doi.org/10.3390/w11071449.

Perini, L., L. Calabrese, G. Salerno, P. Ciavola, and C. Armaroli. 2016. "Evaluation of Coastal Vulnerability to Flooding: Comparison of Two Different Methodologies Adopted by the Emilia-Romagna Region (Italy)." *Natural Hazards and Earth System Sciences* 16 (1): 181–194. https://doi.org/10.5194/nhess-16-181-2016.

Phoon, S. Y., A. Y. Shamseldin, and K. Vairavamoorthy. 2004. "Assessing Impacts of Climate Change on Lake Victoria Basin, Africa." *People-Centred Approaches to Water and Environmental Sanitation: Proceedings of the 30th WEDC Conference 25th - 29th October 2004 Vientiane, Lao PDR*, 392–397.

Sande Van de, Bas, Joost Lansen, and Claartje Hoyng. 2012. "Sensitivity of Coastal Flood Risk Assessments to Digital Elevation Models." *Water* 4 (3): 568–579. https://doi.org/10.3390/w4030568.

Seenath, Avidesh, Matthew Wilson, and Keith Miller. 2016. "Hydrodynamic versus GIS Modelling for Coastal Flood Vulnerability Assessment: Which is Better for Guiding Coastal Management?" *Ocean and Coastal Management* 120:99–109. https://doi.org/10.1016/j.ocecoaman.2015.11.019.

Sene, K. J. 2000. "Theoretical Estimates for the Influence of Lake Victoria on Flows in the Upper White Nile." *Hydrological Sciences Journal* 45 (1): 125–145. https://doi.org/10.1080/02626660009492310.

Stager, J., Curt Alexander Ruzmaikin, Declan Conway, Piet Verburg, and J. Mason. Peter. 2007. "Sunspots, El Niño, and the Levels of Lake Victoria, East Africa." *Journal of Geophysical Research Atmospheres* 112 (15): 1–13. https://doi.org/10.1029/2006JD008362.

Sudong, Xu, and Wenrui Huang. 2011. "Estimating Extreme Water Levels with Long-Term Data by GEV Distribution at Wusong Station Near Shanghai City in Yangtze Estuary." *Ocean Engineering* 38 (2–3): 468–478. https://doi.org/10.1016/j.oceaneng.2010.11.022.

Sutcliffe, John V., and Y. P. Parks. 1999. Salz, H., Kundzewicz, Zbigniew W. eds. The Hydrology of the Nile IAHS Special Publication no. 5 (Wallingford, Oxfordshire, UK, IAHS Press): 1-179. 1-910502-75-9.

Tang, Jianxiong, Li Yanmin, Shenghui Cui, Xu Lilai, Hu Yuanchao, Shengping Ding, and Vilas Nitivattananon. 2021. "Analyzing the Spatiotemporal Dynamics of Flood Risk and Its Driving Factors in a Coastal Watershed of Southeastern China." *Ecological Indicators* 121:107134. https://doi.org/10.1016/j.ecolind.2020.107134.

Tate, Emma, John Sutcliffe, Declan Conway, and Frank Farquharson. 2004. "Water balance of Lake Victoria: update to 2000 and climate change modelling to 2100 / Bilan hydrologique du Lac Victoria: mise à jour jusqu'en 2000 et modélisation des impacts du changement climatique jusqu'en 2100." *Hydrological Sciences Journal* 49 (4): 563–574. https://doi.org/10.1623/hysj.49.4.563.54422.

Tellman, B., J. A. Sullivan, C. Kuhn, A. J. Kettner, C. S. Doyle, G. R. Brakenridge, T. A. Erickson, and D. A. Slayback. 2021. "Satellite Imaging Reveals Increased Proportion of Population Exposed to Floods." *Nature* 596 (7870): 80–86. https://doi.org/10.1038/s41586-021-03695-w.

Temmerman, Stijn, Erik M Horstman, Ken W Krauss, Julia C Mullarney, Ignace Pelckmans, and Ken Schoutens. 2023. "Marshes and Mangroves as Nature-Based Coastal Storm Buffers." *Annual Review of Marine Science* 15 (1): 95–118. https://doi.org/10.1146/annurev-marine-040422-092951.

UBOS. 2016. "National Population and Housing Census 2014 - Main Report." Kampala.

UNDRR. 2020. "Human Cost of Disasters." *Human Cost of Disasters An Overview of the Last 20. Years 2000-2019*. https://doi.org/10.18356/79b92774-en

Wardlaw, R. B., M. Sharif, and F. Kimaite. 2005. "Real-Time Hydro-Power Forecasting on the Victoria Nile." *Proceedings of the Institution of Civil Engineers: Water Management* 158 (2): 45–54. https://doi.org/10.1680/wama.2005.158.2.45.

Wilberforce, Manirakiza, Emmanuel Tumwesigye, Kevin Otim, Mary Akurut, and Harrison E. Mutikanga. 2022. "Lessons Learnt from Dealing with Climatic Extreme Events - a Case of L. Victoria and the White Nile Cascade." *E3S Web of Conferences* 346:04003. https://doi.org/10.1051/e3sconf/202234604003.

World Bank. 2018. "Lake Victoria Environmental Management Project III (P165352). Project Information Document/Integrated Safeguards Data Sheet (PID/ISDS)." World Bank. Washington DC, USA. https://bit.ly/3zzQB8n.

Yin, X., and S. E. Nicholson. 1998. "The Water Balance of Lake Victoria." *Hydrological Sciences Journal* 43 (5): 789–811. https://doi.org/10.1080/02626669809492173.